

Multi-Camera Multi-Vehicle Tracking Dataset and Algorithm Brief

NDNSF-UAV integration note

NDNSF-UAV project

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Purpose. This note describes the public “Multi-Camera Multi-Vehicle Tracking System (UAV Dataset)” and its reference tracking pipeline, and maps them to a bounded NDNSF-UAV experiment. The dataset is useful for a small, reproducible multi-provider prototype; it is not by itself a complete multi-UAV scientific benchmark for recognition or geolocation.

1 Dataset overview and provenance

The dataset is distributed with the project [Hugging Face dataset card](#). The accompanying implementation is available at [the project repository](#). The repository describes three synchronized one-minute UAV clips: `uav1_1min.mp4`, `uav2_1min.mp4`, and `uav3_1min.mp4`. The views observe overlapping urban regions and are intended to maintain a consistent identity for the same vehicle across cameras. The CSV output uses frame-level fields such as bounding box, confidence, local tracker ID, `global_id`, `camera_id`, speed, and status.

Property	Dataset-level interpretation
Views	Three UAV video streams with synchronized frame indices
Target	Vehicles in urban scenes; the example highlights a red vehicle with a persistent global ID
Labels	Per-frame boxes and tracking metadata; local and cross-camera IDs are both represented
Output	Annotated MP4 files and per-camera CSV files
Algorithm context	Ultralytics YOLO detection, Supervision/ByteTrack local tracking, and overlap-region cross-camera matching
Scale	Small pilot resource: three one-minute videos rather than a large, diverse flight campaign
License boundary	Check the dataset card and project license before redistribution; keep raw videos outside Git

The data should therefore be called a *synchronized multi-camera UAV tracking pilot*. It should not be described as a broad recognition benchmark: the release does not provide the target-class diversity, calibrated camera geometry, or flight metadata needed for a general claim about 3-D localization or improved recognition accuracy. It is also different from ordinary UAV MOT: multi-object tracking within one video does not establish identity association across independent views.

2 Reference algorithm

The reference code is an engineering baseline rather than a new tracking algorithm. Its processing chain is:

1. Load one YOLO detector and initialize one local tracker per UAV stream.
2. Process corresponding frame indices in parallel. Each stream performs confidence filtering, non-maximum suppression, and local ByteTrack association.
3. Convert local detections into anchor points and estimate speed and direction using camera-specific pixel-to-meter scale factors.
4. Use configured field-of-view and overlap regions to match compatible local tracks across cameras and assign a persistent `global_id`.
5. Append detection and association records to CSV and render boxes, labels, traces, regions, counters, and time codes to annotated video.

The important distinction is between three identities: the detector's object candidate, a camera-local `tracker_id`, and the cross-camera `global_id`. A global ID is an application-level association result; it is not automatically an NDN producer identity and it is not a cryptographic proof that two objects are physically the same vehicle. The matching logic is based on overlap regions and configured spatial/appearance cues, so its speed and direction estimates require scene-specific calibration.

For an NDN SF experiment, each camera stream should be treated as a separate logical Provider. The application should publish a bounded frame or temporal chunk as Named Data, while the tracking service consumes references to those objects. The full video should not be placed in a Request or Selection message.

3 NDNSF-UAV integration and evaluation

3.1 Data-plane mapping

The three streams can be mapped to a minimal collaboration plan:

```
/uav/A -> ViewData-A --+
/uav/B -> ViewData-B ----> TrackingProvider -> User
/uav/C -> ViewData-C --+
```

The protected discovery descriptor may contain the target region, deadline, minimum number of views, required vehicle-tracking capability, and resource constraints. After ACK collection and Selection, the plan assigns capture roles to the selected UAV Providers and a tracking role to the analysis Provider. Each View Data object should carry at least:

- a request/mission session and attempt identifier;
- producer identity, camera/UAV identity, sequence and frame index;
- capture timestamp, target or track identifier, and media type;
- exact Data name, content digest, bounding-box metadata, and optional pose/GPS information.

The tracker must accept a View Data object only when its producer, name, digest, capture window, and dependency edge agree with the selected plan. Directly retrieved and cached objects must pass the same checks. A missing Provider, wrong producer, undeclared dependency, duplicate object, or stale attempt is a protocol rejection—not a computer-vision false negative.

3.2 Recommended experiment stages

1. **Pipeline smoke test.** Use the three released videos to exercise three ACKs, three selected capture roles, Named Data publication, and one tracking Provider. Verify direct retrieval, cache retrieval, and result publication.
2. **Failure and admission tests.** Remove one stream, delay one stream, substitute a wrong producer, replay an old frame, or submit an undeclared dependency. Record the first rejection stage and the final request outcome.
3. **Paired tracking comparison.** Compare one stream against two and three streams over the same time windows. Keep detector weights, thresholds, frame sampling, and preprocessing fixed.
4. **Extended application study.** Only after obtaining more scenes and target identities should the project report general multi-view tracking or recognition improvements. The current three one-minute release is a pilot, not sufficient evidence for such a claim.

3.3 Metrics and limitations

Application metrics should include IDF1, HOTA or MOTA, identity switches, cross-view association accuracy, track continuity, and recovery after occlusion. NDNSF metrics should include positive ACK count, selected Provider count, accepted-view count, completion within deadline, end-to-end latency, message/byte overhead, cache-hit latency, and rejection counts for wrong producer, wrong dependency, and stale attempt.

The dataset is especially useful for validating the *data-plane contract* of NDNSF-UAV: selected Providers publish named objects and the analysis role advances only on authorized, correctly correlated objects. It does not by itself validate UAV flight control, GPS localization accuracy, cryptographic authorization strength, or that adding views always improves recognition. Those claims require calibrated metadata, more independent scenes, target-level splits, and a separately controlled application study.

Sources

- Dataset card: [Hugging Face README](#)
- Algorithm repository: <https://github.com/JYe9/multi-camera-multi-vehicle-tracking-system>
- Project paper record: <https://arxiv.org/abs/2605.15779>